

Industrial LPG Bulk Storage & Plant Safety Norms

BANGLADESH LPG SAFETY & REGULATORY COMPLIANCE PLATFORM

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Mandatory Engineering Norms for Bulk Storage Tanks, Manifolds & Auto-Shutoff

Statutory Reference: Department of Explosives (DoE) & Bangladesh Energy Regulatory Commission (BERC)

1. BULK STORAGE SAFETY CLEARANCES (NFPA 58 / DoE RULE 1991)

- [*] Maintain minimum 15-meter buffer from property boundaries for storage > 20,000L.
- [*] Install high-decibel hydrocarbon vapor sensors at 0.3m above ground near valves.
- [*] Ensure dual earthing electrodes with ground resistance < 1.0 Ohm across all vessels.
- [*] Equip all discharge manifolds with fail-safe pneumatic Emergency Shutoff Valves (ESV).

2. PERIODIC INSPECTION, TESTING & STATUTORY LICENSING

- [*] Perform mandatory hydrostatic pressure testing every 5 years under DoE surveyor.
- [*] Calibrate pressure relief valves (PRV) annually; set pressure at 1.1x design rating.
- [*] Maintain 24/7 dedicated deluge water spray system supplying 10.2 L/min/m2.
- [*] Renew Department of Explosives (DoE) and Fire Service licenses prior to expiration.

3. PLANT OPERATIONAL PROTOCOLS & ACCIDENT PREVENTION

- [*] Prohibit all non-intrinsically safe electrical equipment inside Zone 1 & Zone 2 areas.
- [*] Enforce strict Permit-to-Work (PTW) protocols for all hot work within 30 meters.
- [*] Conduct quarterly full-scale emergency evacuation drills with local Fire Service.

24/7 LPG EMERGENCY HOTLINES: Fire Service: 16137 | National Emergency: 999 | DoE Central: +880 2-9335445

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